

Software Requirements Specification (SRS) & Work Breakdown Structure (WBS)

Problem Statement #42: Internal Microservice Catalog & Health Portal | Student: Subramani B M (PES1UG24CS473) | Section H

1. System Scope & Functional Requirements

ID	Title	Specification	Acceptance Criteria
FR-001	Automated Health Monitoring	Automated 30-sec health probes; computes rolling 24h availability.	Pass: Downtime recorded on 3 failed probes. Fail: Down service marked healthy.
FR-002	Service Catalog & Registry	Searchable catalog of registered services with schemas, owner, repo URL.	Pass: Service indexed and discoverable in <60s. Fail: Stale metadata.
FR-003	Dependency Mapping	Dynamic graph visualizing upstream/downstream relationships for 200+ services.	Pass: Sub-2.0s render SLA; blast radius visualizer. Fail: Missing links.
FR-004	Multi-Channel Alerting	Real-time downtime alert dispatch via Slack, PagerDuty, and SMTP.	Pass: Alert delivered to on-call within 60s. Fail: Missed alert.
FR-005	API Doc Aggregator	Auto-fetches OpenAPI 3.0 / Swagger specs with mock sandbox runner.	Pass: Schema rendered in <2 min. Fail: Broken/missing schema.
NFR-001	Performance SLA	Graph renders up to 200 nodes smoothly in ≤2.0s on standard desktop browsers.	Pass: Latency ≤2.0s under 50 concurrent users. Fail: >2s latency.
NFR-002	Security & Compliance	TLS 1.2+ encryption, enterprise SSO (OAuth2/SAML), HashiCorp Vault secrets.	Pass: Zero plaintext credentials; strict RBAC. Fail: Auth bypass.

2. Work Breakdown Structure (WBS) & Jira Task Mapping

WBS	Work Package Title	Jira Key	Points	Core Engineering Deliverable
1.1.1	API Gateway SSO / OAuth2 Auth	SBP42-1	5 SP	TLS 1.2+ ingress filter validating JWT auth tokens.
1.1.3	HashiCorp Vault Secret Client	SBP42-2	3 SP	Secure storage isolating Slack tokens & probe credentials.
1.2.1	Service Registration REST API	KBPS42-1	5 SP	CRUD endpoints for service onboarding & schema validation.
1.2.2	PostgreSQL Metadata Datastore	KBPS42-2	3 SP	Relational database schema with audit tables.
1.3.1	30s Health Pinger Engine	SBP42-3	8 SP	Async worker polling 200+ microservices every 30s.
1.3.2	TimescaleDB Time-Series Engine	SBP42-4	5 SP	Hypertables storing timestamped latency & uptime.
1.4.1	Neo4j Graph Topology Modeler	SBP42-5	8 SP	Directed graph store for upstream/downstream dependencies.
1.4.2	Sub-2.0s Graph UI Renderer	SBP42-6	13 SP	Optimized front-end canvas fulfilling NFR-001 render SLA.
1.5.1	RabbitMQ Event Bus Messaging	SBP42-7	5 SP	AMQP exchange buffering outage.detected events.
1.5.3	Multi-Channel Alert Dispatcher	KBPS42-5	5 SP	Slack, PagerDuty & SMTP dispatchers (<60s SLA).
1.6.1	OpenAPI 3.0 Ingestion Pipeline	KBPS42-6	5 SP	Swagger parser converting JSON specs into interactive docs.
1.7.1	End-to-End SLA Benchmarking	SBP42-8	8 SP	Complete verification validating all FRs, NFRs, and RTM.